TECHNOLOGICAL CHINA

From the world's factory
to a technological powerhouse

INDUSTRIAL TRANSFORMATION

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CONCEPTUAL ILLUSTRATION

THE CENTRAL QUESTION

How does a technological powerhouse emerge?

The answer cannot be reduced to one invention, one company, or one policy.

01

INVENT

Discover, design, and develop.

02

MANUFACTURE

Turn ideas into reliable products.

03

LEARN

Improve with every production cycle.

04

COORDINATE

Connect talent, suppliers, and capital.

ROAD MAP

Five questions about industrial transformation

History, industrial case studies, and exercises to examine underlying mechanisms.

01	Where does industrial capability come from?	History + UNIDO
02	How does an advanced factory operate?	Shougang + Midea
03	Why do robots matter?	IFR + Unitree + Xiaomi
04	What is China pursuing by 2030?	AI+ + 15th Plan
05	Where do the limitations emerge?	Chips + demand + exports

SECTION 01

HISTORY

Before AI, there was industry

Today's advantage rests on decades of investment, learning, and production networks.



INDUSTRIAL TRAJECTORY

Four stages, one continuous accumulation of capabilities

Teaching periodization: stages overlap and do not imply abrupt transitions.

1978–2001

Reform and opening up

Investment, special economic zones, initial capabilities

2001–2014

Global integration

Exports, logistics, suppliers, and learning

2015–2025

Industrial modernization

Automation, R&D, batteries, and equipment

2026–2030

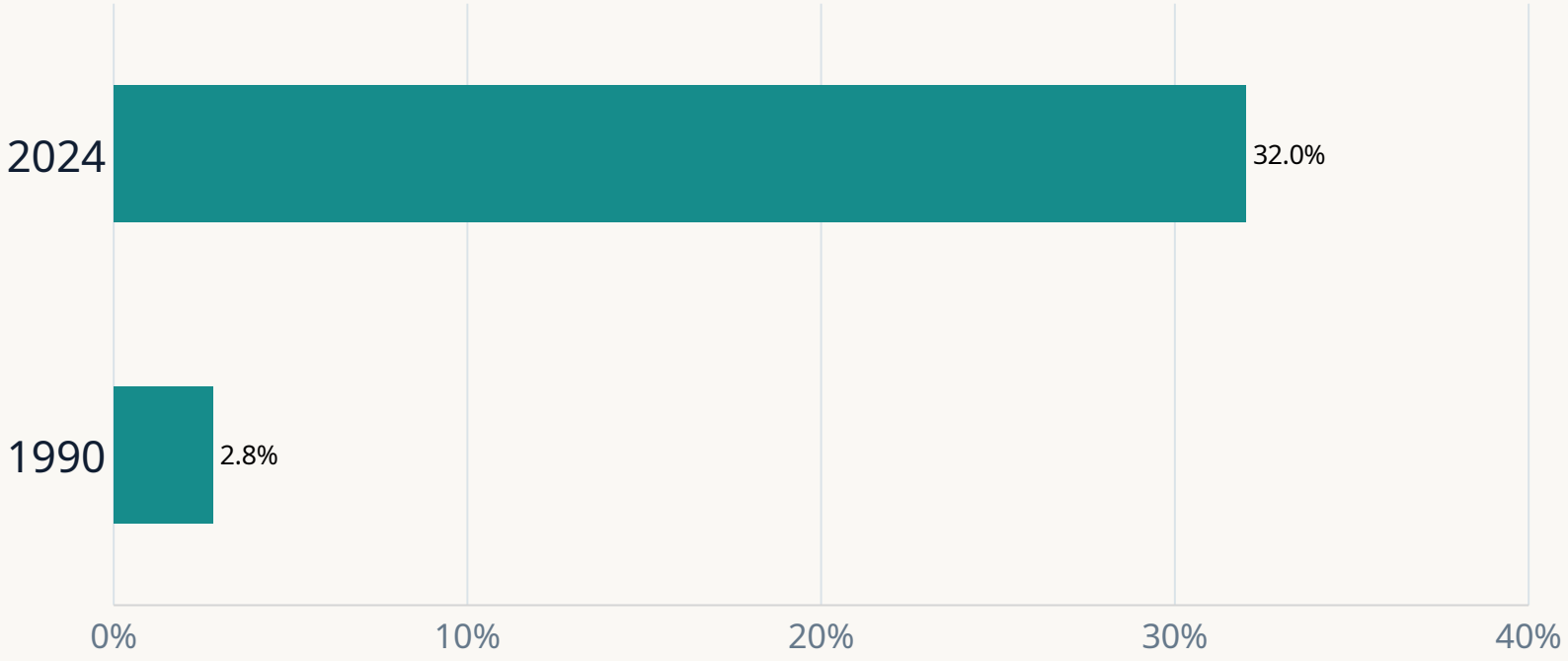
Intelligent convergence

AI+, industrial AI, robotics, and selective autonomy

THE GREAT SHIFT

From less than 3% to nearly one-third of the world

China's share of global manufacturing value added.



×11.4

increase in the share between the two years

This share measures manufacturing value added, not productivity per worker.

INDUSTRIAL CAPABILITIES

A factory's strength depends on its ecosystem

Knowledge is distributed among firms and organizations.



Connections enable the sharing of knowledge, resources, and responsiveness.

TEACHING EXERCISE 01

Two companies design the same robotic hand

Hypothetical example illustrating supplier coordination times.

COMPANY A

Dispersed suppliers

20 weeks

COMPANY B

Integrated supplier network

8 weeks

The difference is not “innate intelligence”: it is coordination, prototypes, learning, and logistics.

Planning, local experimentation, and competition

Central coordination coexists with provincial experimentation and business competition.

CENTRAL GOVERNMENT

Plans, standards, priorities

LOCAL GOVERNMENTS

Industrial parks, infrastructure, incentives

COMPANIES

Design, manufacturing, competition

SUPPLIERS

Components, lead times, improvements

POTENTIAL ADVANTAGE: learning / RISK: overinvestment and duplication

INDUSTRIAL GEOGRAPHY

China is a network of technological hubs

Specializations are not exclusive: they overlap and evolve.

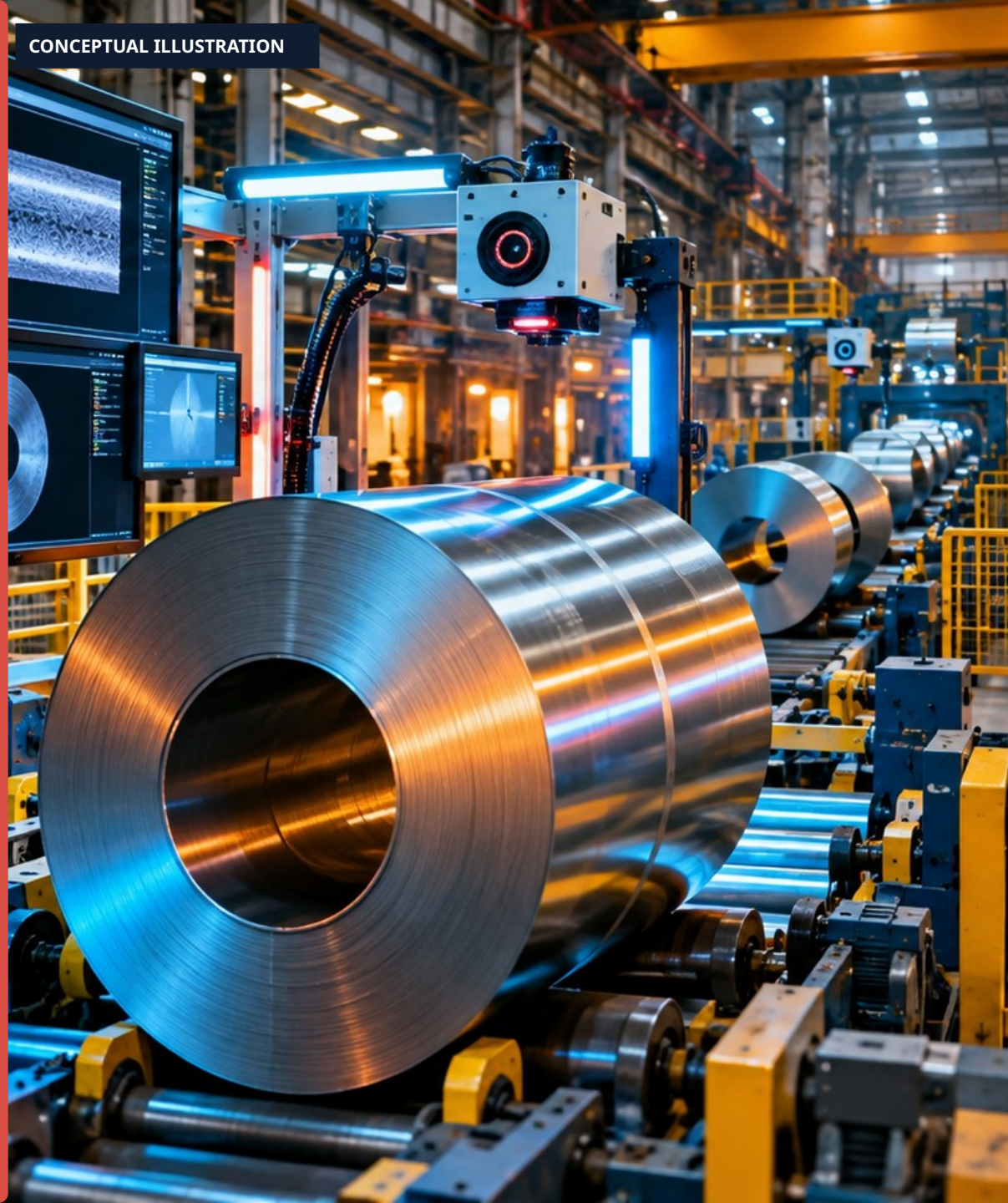
BEIJING	Universities, research, AI
SHANGHAI / SUZHOU	Advanced manufacturing and equipment
SHENZHEN / GUANGDONG	Electronics, hardware, and vehicles
HANGZHOU	Digital platforms and software
HEFEI / WUHAN / CHENGDU	Specialized industrial hubs

SECTION 02

THE SMART FACTORY

A machine is not a factory

Digitalization, automation, and AI are distinct layers of industrial capability.



THREE DISTINCT CONCEPTS

Digitalization, automation, and artificial intelligence

A factory can do one of these without necessarily doing the others.

01

DIGITALIZE

Record and connect information

Sensors, data, traceability

02

AUTOMATE

Perform tasks under automatic control

PLCs, CNC, robots, actuators

03

USE AI

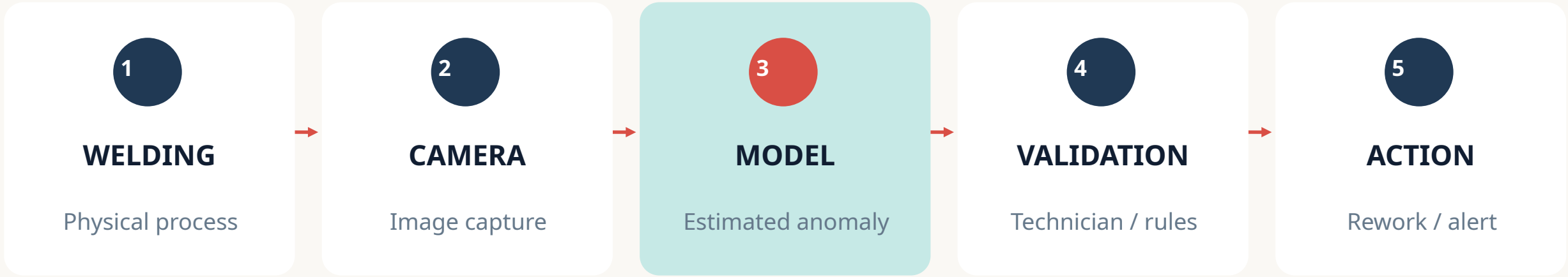
Infer patterns and support decisions

Computer vision, prediction, optimization

DECISION CHAIN

From a weld to a verifiable action

AI helps detect problems; control, engineering, and safety remain essential.



A model cannot “see” every defect or bypass quality validation.

REAL-WORLD CASE · SHOUGANG

Can AI help produce higher-quality steel?

Beijing Shougang Cold Rolling · Global Lighthouse Network.



67

advanced manufacturing technology applications

61%

of them incorporated AI

TRADITIONAL INDUSTRY CASE

SHOUGANG · MEASURED IMPACT

Fewer defects; greater production efficiency

Figures reported by the WEF case study, not a controlled causal experiment.

−35%
Product defects

+21.2%
Production-line efficiency

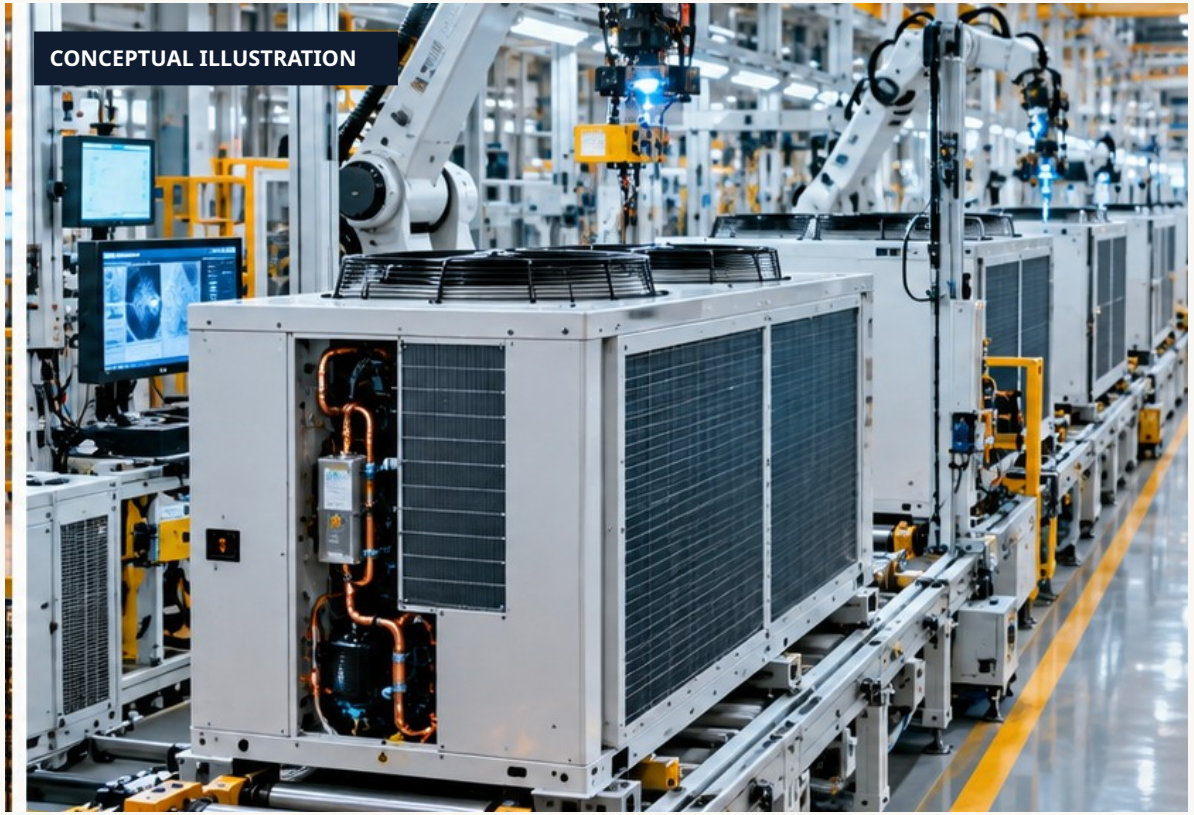
−55%
Customer complaints

If there were 100 defects before, a 35% reduction would leave 65.

REAL-WORLD CASE · MIDEA

A factory that learns to produce different orders

Midea General Refrigeration, Chongqing · Refrigeration and industrial equipment.



FLEXIBLE PRODUCTION

79 digital use cases deployed

Customized orders:

from 31% to 87%

MIDEA · RESULTS

Customization without sacrificing speed or quality

WEF case study: indicators reported following the plant's digital transformation.

−81%
Setup time

−45%
Design time

−31%
After-sales failures

If setup took 100 minutes, an 81% reduction would bring it down to 19 minutes.

REAL-WORLD ROBOTICS · IFR

China accounts for 59% of new robot installations

Industrial robot installations in 2025, a different category from humanoids.

354,000

Industrial robots installed in China

59%

Share of global installations

55% share held by Chinese suppliers in their domestic market

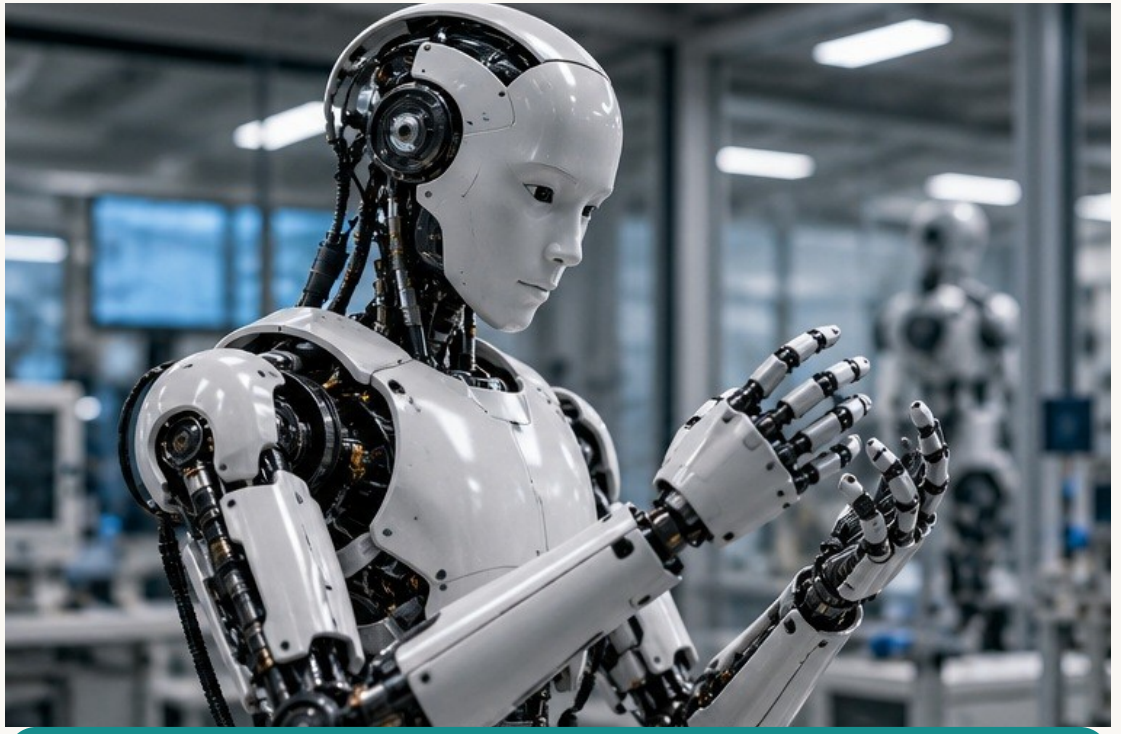
TWO KINDS OF ROBOTICS

An industrial arm and a humanoid are not equivalent

Humanoids are visually compelling; evidence of mass industrial adoption is different.



INDUSTRIAL ROBOT · repeatable task

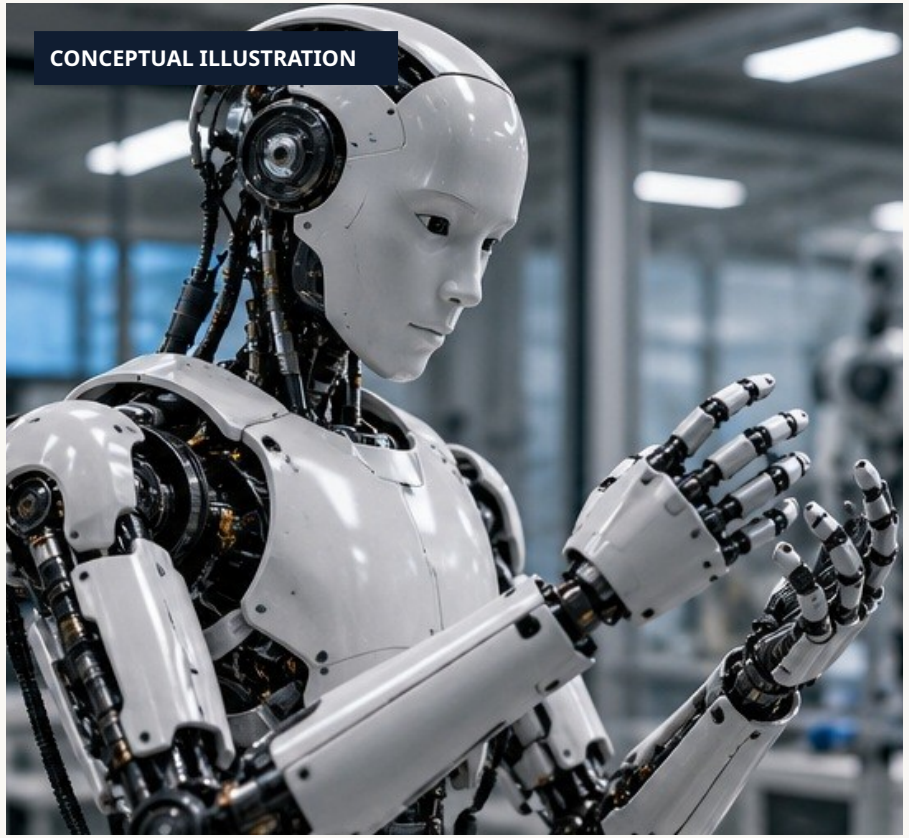


HUMANOID · mobility and manipulation

UNITREE CASE · G1

A humanoid embodies an entire industrial network

Behind its movements lie electronics, motors, control, sensors, and batteries.



ACTUATORS	Motors and transmission
PERCEPTION	Cameras and sensors
COMPUTING	Chips and software
ENERGY	Batteries and management
CONTROL	Motion and balance

UNITREE · EVIDENCE

An impressive video does not prove industrial autonomy

Three levels of evidence that must not be confused.

01	**DEMONSTRATION**	Walks, dances, or performs a routine.	Teleoperated? How many attempts?
02	**TECHNICAL CAPABILITY**	Perceives, controls joints, manipulates objects.	With what precision and in what environment?
03	**PRODUCTIVE VALUE**	Works safely, reliably, and at competitive cost.	What are its uptime, failure rate, and cost?

TEACHING EXERCISE 02

The same 20 robots, different outcomes

Assumptions: same workforce, shifts, complexity, and required quality.

FACTORY A

100 workers + 20 robots

1,000 parts · 5% defects

950

good parts per day

FACTORY B

100 workers + 20 robots

1,150 parts · 2% defects

1,127

good parts per day

Factory B produces 177 more good parts per day: +18.6% without buying more robots.

THE ECONOMIC LOGIC

More robots do not automatically mean higher productivity

Five conditions must be evaluated before investing.

01	**CAPITAL**	Purchase price and installation
02	**OPERATIONS**	Uptime and failures
03	**PEOPLE**	Engineering and training
04	**QUALITY**	Defects and rework
05	**DEMAND**	Sales needed to recoup investment

CASE STUDY · XIAOMI

From electronics to the electric car

A case of capability transfer: software, hardware, suppliers, and engineering.



BEFORE

Electronics and connected devices

CAPABILITY TRANSFER

Design, software, supply chain

NEW SECTOR

Electric vehicles and integrated manufacturing

THE ELECTRIC VEHICLE ECOSYSTEM

China turns industrial capabilities into a new sector

Electric vehicles in China, BEVs and PHEVs. Data for 2025.

≈ 16 M

Electric vehicles manufactured in China

≈ 75%

Share of global production, 2025

>2.5 M

China's electric vehicle exports in 2025

Batteries + motors + chips + software + suppliers + market = integrated industry

SECTION 03

AI + INDUSTRY

Artificial intelligence leaves the laboratory

Its value depends on data, people, and processes capable of putting it to work.



OFFICIAL CONCEPT · AI+

Six areas of application for artificial intelligence

China's State Council · Guidelines issued in August 2025.

1 **SCIENCE**

Discovery and simulation

2 **INDUSTRY**

Manufacturing and processes

3 **CONSUMPTION**

Products and services

4 **PUBLIC WELL-BEING**

Health and education

5 **GOVERNANCE**

Public administration

6 **COOPERATION**

International cooperation

AI+ · TARGETS

The AI+ targets for 2027 and 2030

Target penetration of next-generation smart terminals and AI agent applications.

>70%

Official target for 2027

>90%

Official target for 2030

2035

Horizon for an intelligent economy and society

The indicator refers to smart terminals and AI agent applications.

TEACHING EXERCISE 03

How can AI help prevent a machine from stopping?

HYPOTHETICAL example of predictive maintenance.

BEFORE

160 h

of unplanned downtime per year

AFTER

80 h

of unplanned downtime per year

50% less downtime does NOT mean 50% more total production.

CONDITIONS FOR USE

The requirements for industrial AI adoption

Buying a model is not the same as improving a factory.

DATA

Are the sensors reliable?

MODELS

Do they solve a real problem?

PEOPLE

Who validates and repairs?

PROCESSES

Can the decision actually be implemented?

Integration determines how effectively technology is used.

SECTION 04

STRATEGY 2030

The 15th Five-Year Plan: new productive capabilities

The objective is to reshape the industry's technology mix, not merely increase its output.



THREE POLICY HORIZONS

Three instruments of industrial modernization

Distinct instruments that overlap and respond to changing needs.

2015

MADE IN CHINA 2025

Modernize industries, equipment, and components.

2025

ARTIFICIAL INTELLIGENCE+

Spread AI across many sectors of the economy.

2026–2030

15TH FIVE-YEAR PLAN

Integrate innovation, industry, and technological security.

A NEW DEVELOPMENT CONCEPT

“New quality productive forces” 新质生产力

Three industrial horizons under the 15th Five-Year Plan.

01 **UPGRADE EXISTING INDUSTRIES**

Steel, machinery, chemicals: quality and efficiency.

02 **SCALE UP EMERGING INDUSTRIES**

Robotics, smart vehicles, new materials.

03 **EXPLORE FUTURE INDUSTRIES**

Embodied AI, quantum technologies, 6G, and more.

FROM POLICY DECISIONS TO DATA

A plan is not an operational factory

How to assess an industrial program at four different levels.

01

ANNOUNCEMENT

Goals and priorities

02

POLICY INSTRUMENT

Rules, budgets, incentives

03

IMPLEMENTATION

Plants, contracts, training

04

OUTCOME

Productivity, quality, sales

Four distinct levels: an official target does not prove that an investment was profitable.

SECTION 05

GEOPOLITICS

The network also has weak points

Manufacturing strength can coexist with dependence, overinvestment, and vulnerability.



TEACHING EXERCISE 04

One missing part can stop the entire line

Hypothetical example: one essential chip per device, with no substitute.

10,000
Weekly factory capacity

6,000
Critical chips available

Maximum possible output: 6,000 devices → 40% of capacity unused

TECHNOLOGICAL RIVALRY

China and the United States have different strengths

Competition depends on the segment analyzed, not one overall scoreboard.

CHINA

- Manufacturing scale
- Dense supplier networks
- Industrial robot installations
- Batteries and electric vehicles

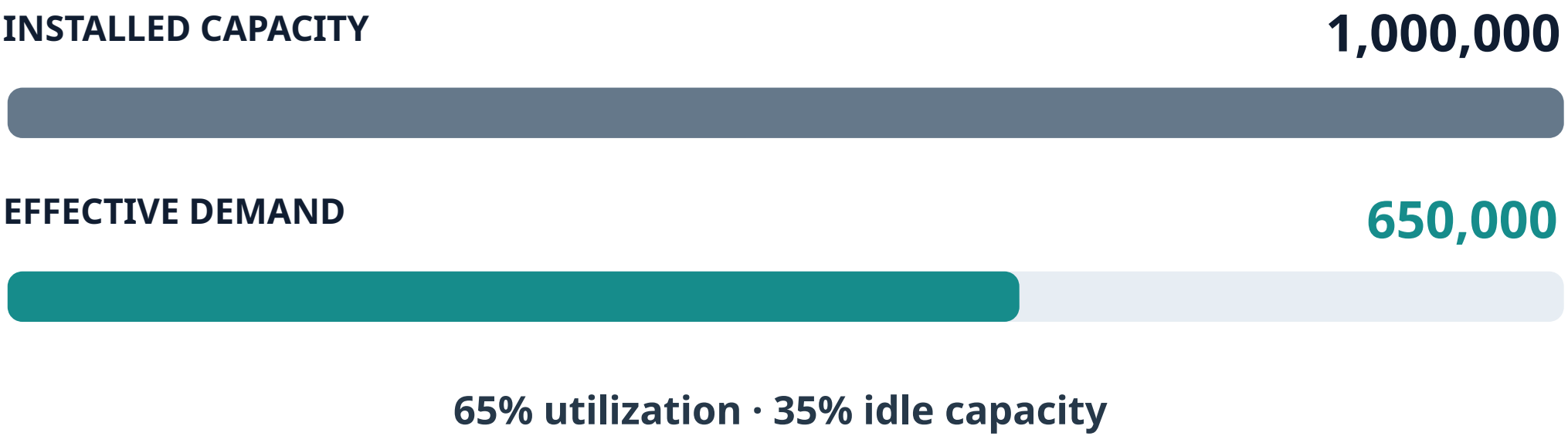
UNITED STATES

- Frontier-model ecosystem
- Cloud computing and digital platforms
- Specialized software and tools
- International technology alliances

THE PARADOX OF SUCCESS

A strong industry can also generate excess capacity

Hypothetical example of installed capacity and demand in comparable units.



REAL-WORLD CASE · ELECTRIC VEHICLES

More production, intense competition, and rising exports

Electric vehicles in 2025. Production, sales, and exports are different series.

≈16 M

EVs manufactured in China

>13 M

Domestic sales in 2025

>2.5 M

Exports from China in 2025

Price competition squeezes margins: exports can help sustain sales.

A RIGOROUS READING

Four common confusions to avoid

Government, corporate, and teaching data have different evidentiary weight.

01	**ANNOUNCEMENT ≠ ACHIEVEMENT**	A plan specifies targets, not results.
02	**DEMO ≠ AUTONOMY**	A walking robot does not prove profitable work.
03	**ROBOTS INSTALLED ≠ MANUFACTURED**	Adoption and production are different measures.
04	**REPORTED CASE ≠ CAUSALITY**	Improvement may reflect several technologies.

THE CENTRAL THESIS

How did China build a technological powerhouse?

A sequence of accumulated capabilities, not a single-cause explanation.

01

HISTORY

Build the capacity to manufacture

→

02

NETWORKS

Connect firms and knowledge

→

03

LEARNING

Improve design and processes

→

04

TECHNOLOGY

Integrate robots, data, and AI

→

05

OUTCOMES

Compete, innovate, and face limits

Innovation is built through manufacturing, coordination, and learning. It is not necessarily guaranteed.

THREE QUESTIONS FOR DISCUSSION

What we still need to investigate

Three perspectives on China's technological future.

1. **How much of China's advantage comes from scale, and how much from innovation of its own?**

2. **When does a robot improve productivity, and when does it simply add costs?**

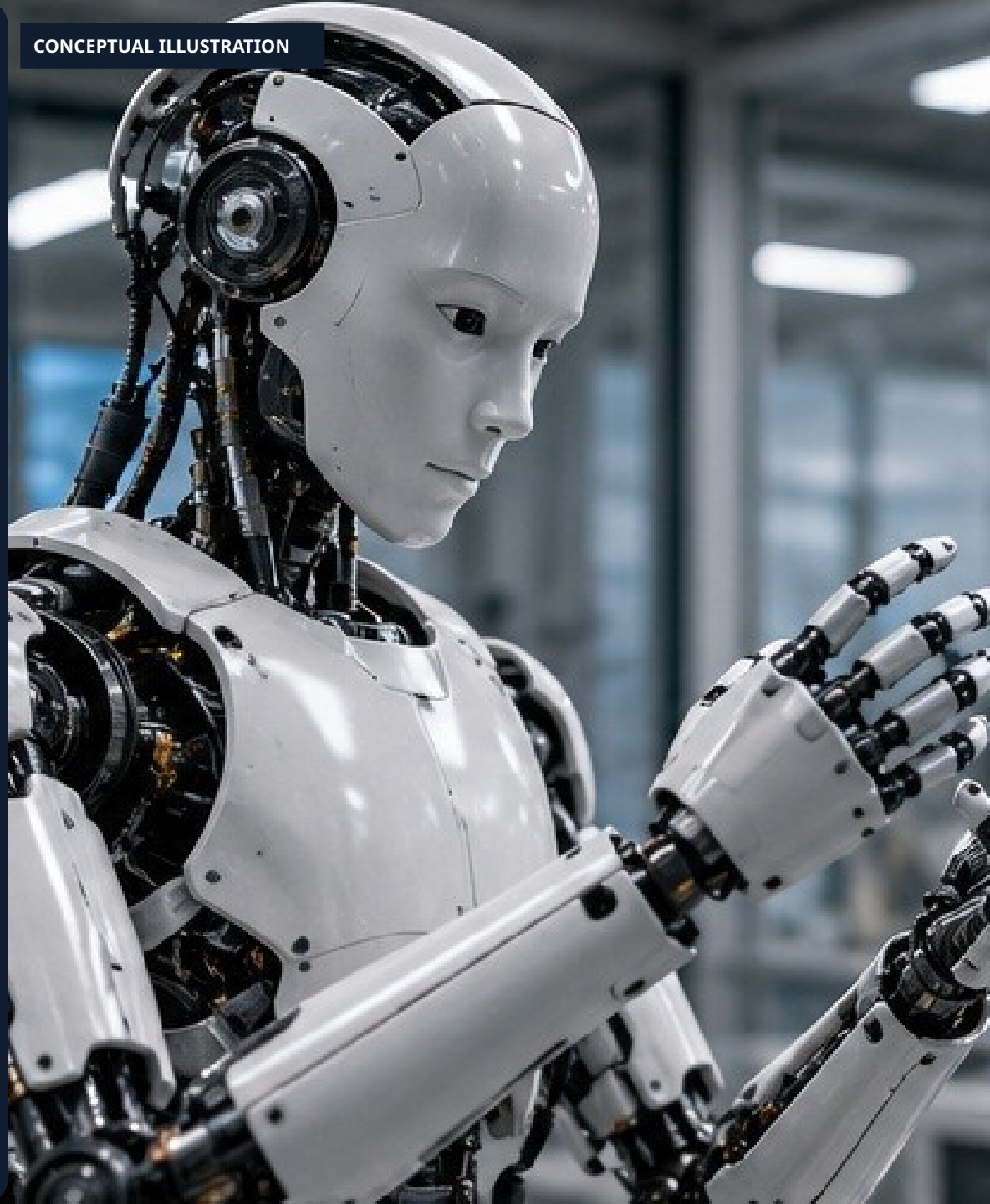
3. **Can industrial leadership exist without full autonomy in advanced chips?**

AI MODELS AND INTELLIGENT ROBOTS

FROM THE INDUSTRIAL SYSTEM TO EMBODIED INTELLIGENCE

DeepSeek · Qwen · agents · Unitree · UBTech

What can models and robots actually do?



SOURCES · I

Statistics and international organizations

Selected references for verifying the main indicators.

UNIDO · International Yearbook of Industrial Statistics 2025
Global share of manufacturing value added.

IFR · World Robotics 2026
Industrial robot installations in China and global share.

IFR · China and the United States, World Robotics 2026
2025 installations and the distinction between adoption and manufacturing.

IEA · Global EV Outlook 2026
Production, sales, exports, and batteries.

Stanford HAI · AI Index 2026
International comparison of AI development.

SOURCES · II

Corporate cases and manufacturing studies

Reported outcomes must be attributed to the sources presenting them.

World Economic Forum · Global Lighthouse Network 2025
Shougang: 67 applications, defects −35%, efficiency +21.2%.

World Economic Forum · Global Lighthouse Network 2025
Midea Chongqing: 79 digital use cases, setup time −81%.

Unitree Robotics · Official G1 specifications
Components and technical capabilities claimed by the manufacturer.

Xiaomi Auto · Hyperfactory
Company description of manufacturing and processes.

IEA · Manufacturing and Trade 2026
The electric vehicle value chain and its economic tensions.

Official documents, economics, and teaching resources

Policy texts and references for further reading after class.

State Council · AI+ Guidelines (2025) ↗
Targets and six areas of action. Official Chinese text.

CSET Georgetown · Technical translation of AI+ ↗
Reference for interpreting official terminology.

NDRC · 15th Five-Year Plan, 2026–2030 ↗
Industrial modernization strategy and future industries.

World Bank · China Economic Update 2026 ↗
Demand, structural reforms, and growth.

Instructor's simulator · Robotic capital and learning ↗
Teaching experiment, not an empirical estimate.

THANK YOU

Understanding capabilities, mechanisms, and limitations

Material prepared by Professor Sergio Gevatschnaider

